

Name: _____

SOLUTIONS

Date: _____



Year 9 Maths

Investigation 2, 2012

Topic – Space and Measurement

In class component ONLY

Total Time: 45 minutes
Reading Time: 5 minutes
Working Time: 40 minutes
Weighting: 4% of the year.

Equipment: Calculator

Important Information:

This investigation will be completed under test conditions. Answers should be rounded appropriately. All working should be shown in the space provided. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks.

No pen, pencils, highlights etc. may be used during reading time. This time is to be used to read through the assessment and check that you understand what is being asked of you. You may speak with the teacher/supervisor during this time (by putting up your hand and waiting patiently for them to approach you) but you may only ask clarification questions and not how to solve the problems. After reading time has ended, you may not ask any more questions.

AIM: In this assessment, you will be investigating the perimeter of figures made from 1cm squares..

Marks Summary

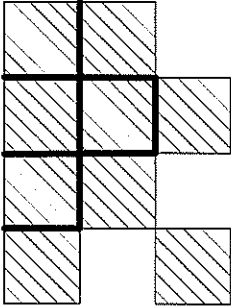
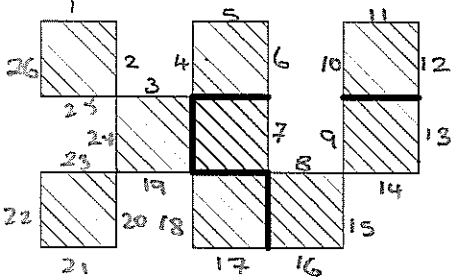
Task	Marks available	Marks awarded
1	2	
2	14	
3	3	
4	5	
5	5	
6	12	
Bonus	1	
Part 2	42	

$$\frac{\quad}{42}$$

$$= \frac{\quad}{\quad} \%$$

BONUS MARKS (look at the whole assessment):

☐ Units: appropriate (+1). If inconsistent (+1/2 only)

Task 1: [2 Marks]	These diagrams show two arrangements of nine squares, which touch either along a side or on a corner. If each square has sides of 1cm, Shape A has 9 shared sides and the distance around (perimeter of) shape A is 18 cm.	A) 
	Shape B has five shared sides. Show how the distance around the shape is calculated to be 26 cm.	B) <u>EXAMPLE</u>  1 MARK MAY BE AWARDED FOR A REASONABLE ATTEMPT BUT WITH ONE OR TWO ERRORS

[2]

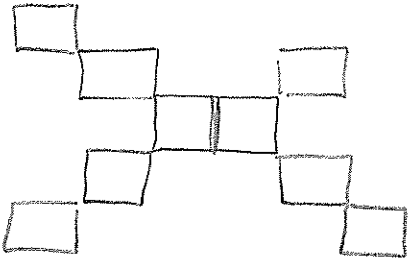
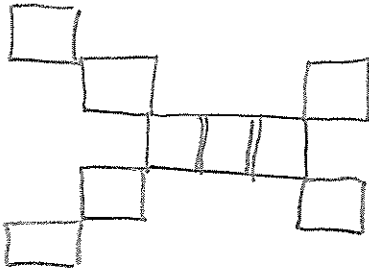
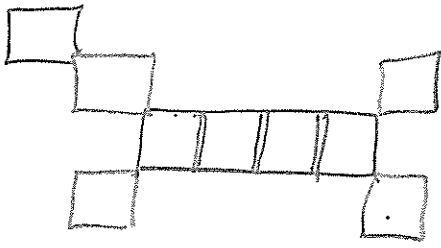
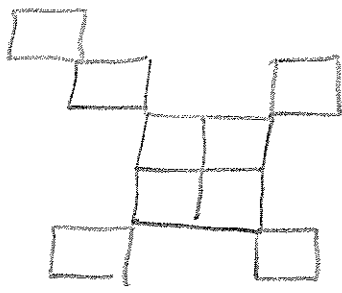
Task 2: [14 Marks]

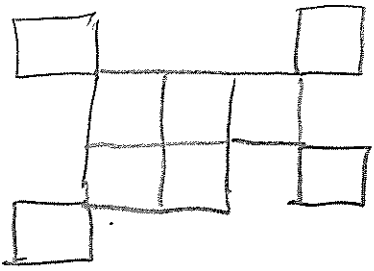
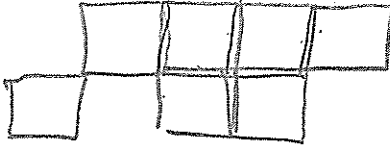
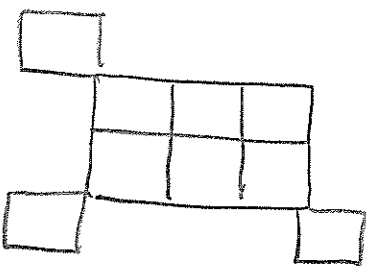
Draw 7 other arrangements of nine squares that are different to the ones above and different to each other.

(Note: The squares must be connected in different ways. Rotated or flipped versions of a shape are not permitted.

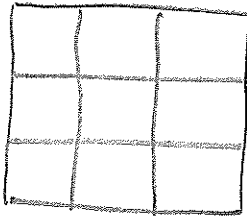
These will be labelled 1 to 7.)

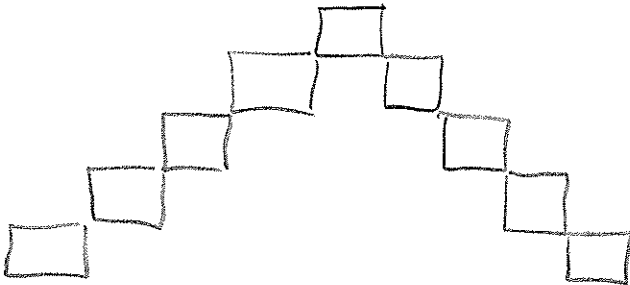
When you have drawn the shapes, calculate the perimeter and record the perimeter and number of shared sides in the appropriate place in the table in Task 3.

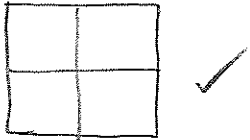
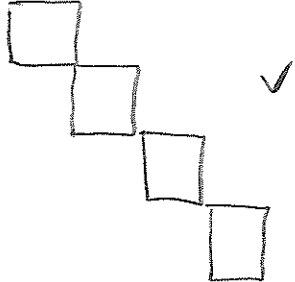
	Shape 1: One shared side Perimeter: <u>34 cm</u> [1]	Drawing:  [1]
	Shape 2: Two shared sides Perimeter: <u>32 cm</u> [1]	Drawing:  [1]
	Shape 3: Three shared sides Perimeter: <u>30 cm</u> [1]	Drawing:  [1]
	Shape 4: Four shared sides Perimeter: <u>28 cm</u> [1]	Drawing:  [1]

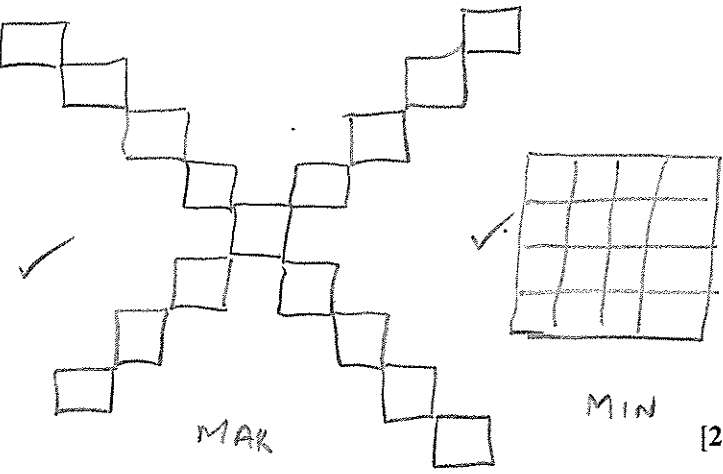
	Shape 5: Five shared sides Perimeter: <u>26 cm</u> [1]	Drawing:  [1]
	Shape 6: Six shared sides Perimeter: <u>24 cm</u> [1]	Drawing:  [1]
	Shape 7: Seven shared sides Perimeter: <u>22 cm</u> [1]	Drawing:  [1]

Task 3	Summary table for nine squares [3 Marks: 1, 2]																																
a)	<table><tr><th>Shape</th><th>Number of shared sides</th><th>Perimeter (cm)</th></tr><tr><td>A</td><td>9</td><td>18</td></tr><tr><td>B</td><td>5</td><td>26</td></tr><tr><td>1</td><td>1</td><td>34</td></tr><tr><td>2</td><td>2</td><td>32</td></tr><tr><td>3</td><td>3</td><td>30</td></tr><tr><td>4</td><td>4</td><td>28</td></tr><tr><td>5</td><td>5</td><td>26</td></tr><tr><td>6</td><td>6</td><td>24</td></tr><tr><td>7</td><td>7</td><td>22</td></tr></table>			Shape	Number of shared sides	Perimeter (cm)	A	9	18	B	5	26	1	1	34	2	2	32	3	3	30	4	4	28	5	5	26	6	6	24	7	7	22
Shape	Number of shared sides	Perimeter (cm)																															
A	9	18																															
B	5	26																															
1	1	34																															
2	2	32																															
3	3	30																															
4	4	28																															
5	5	26																															
6	6	24																															
7	7	22																															
	ALL COPIED CORRECTLY FOR THE MARK [1]																																
b)	In what way does the number of shared sides affect the size of the perimeter PERIMETER = 36 - 2x SHARED SIDES cm ✓✓ (OR EQUIVALENT) <u>OR</u> EVERYTIME YOU SHARE A SIDE THE PERIMETER IS REDUCED BY 2cm ✓																																
	[2]																																

Task 4: [5 Marks: 2, 1, 2]		
	The minimum perimeter around any of these nine square arrangements is 12 cm	
a)	Draw the shape that gives this minimum perimeter.	 [2]
b)	Describe what type of shape gives this minimum perimeter.	SQUARE [1]
c)	Explain why you think this shape might give the minimum perimeter.	MAXIMUM NUMBER OF SHARED SIDES [2]

Task 5: [5 Marks: 2, 1, 2]		
	The maximum perimeter around any of these nine square arrangements is 36 cm	
a)	Draw the shape that gives this <i>maximum</i> perimeter.	 [2]
b)	Describe what type of shape gives this maximum perimeter.	ALL SQUARES JOINED AT CORNERS ONLY [1]
c)	Explain why you think this shape might give the maximum perimeter.	NO SHARED SIDES [2]

Task 6:	[12 Marks: 6, 6]	
	Use your solutions in Tasks 4 and 5 to draw and determine the minimum and maximum perimeter for each of the following.	
a)	A shape with 4 squares instead of 9.	Drawings  MINIMUM  MAXIMUM [2] Minimum perimeter 8 cm [2] Maximum perimeter 16 cm [2]

b)	A shape with 16 squares.	Drawings  Minimum perimeter 16 cm Maximum perimeter 64 cm [2] [2]
----	--------------------------	--